

Department of Computer Science & Engineering

Project Proposal for VII Semester Session 2026-27

Title of the Project:-

METRIX – Online Verification and Digital Certification System for Weighing and Measuring Instruments

SIH Problem No. and Statement:-

SIH26036 – Development of an Online Verification System for Weighing and Measuring Instruments

Name & Designation of the Supervisor/Guide:-

Name: _____

Designation: _____

Project Summary (Max. 150 words) :-

The proposed project, METRIX, is a secure web-based Online Verification and Digital Certification System for weighing and measuring instruments. The system aims to digitize the processes involved in instrument registration, verification, re-verification, scheduling, inspection recording, certificate generation, and validity monitoring. Stakeholders will be able to register instruments and submit verification applications online, while authorized Legal Metrology Officers (LMOs) and Government Approved Test Centres (GATCs) will be able to review applications, schedule verification activities, record inspection observations, and issue digital verification certificates. The system will generate QR-enabled certificates that can be authenticated through a public verification portal. It will also provide role-based dashboards, search and retrieval facilities, document uploads, expiry tracking, automated reminders, and reporting capabilities. Role-based authentication and secure data management will restrict access to authorized users. The project will provide a centralized web platform for managing the verification lifecycle of weighing and measuring instruments.

Technical Details of the Project:-

1. Origin of the Proposal (Motivation): -

The verification of weighing and measuring instruments involves several activities, including application submission, scheduling, inspection, recording of observations, certificate issuance, and monitoring of certificate validity. The SIH26036 problem statement identifies the need to reduce manual processes and improve the management and accessibility of verification records. It also identifies the need for digital verification certificates, centralized databases, and online access to verification history.

- Reduce dependency on manual paperwork.
- Centralize instrument and verification records.
- Enable online submission and tracking of applications.
- Simplify verification scheduling and allocation.
- Digitally record verification observations and results.
- Generate QR-enabled digital certificates.
- Monitor certificate validity and expiry dates.

- Provide automated reminders for re-verification.
- Improve retrieval and accessibility of verification records.

2. Definition of the Problem:-

Verification and re-verification of weighing and measuring instruments involve multiple activities performed by different stakeholders. Manual or disconnected record-keeping can make it difficult to efficiently manage applications, schedule verification activities, maintain instrument history, issue certificates, and monitor certificate validity.

The proposed project addresses this problem by developing a unified web-based verification and certification platform that manages the instrument lifecycle from registration and application submission to scheduling, verification, certificate generation, and re-verification.

The platform will provide role-based functionality for stakeholders, Legal Metrology Officers (LMOs), Government Approved Test Centres (GATCs), and administrators.

3. Objectives:-

1. To develop a secure web-based platform for verification and certification of weighing and measuring instruments.
2. To provide online registration and profile management for stakeholders.
3. To enable online submission of verification and re-verification applications.
4. To provide application tracking and workflow management.
5. To facilitate scheduling and allocation of verification activities.
6. To digitally record instrument specifications, inspection observations, and verification results.
7. To generate QR-enabled digital verification certificates.
8. To provide a public facility for QR-based certificate authentication.
9. To maintain centralized records of instruments, applications, verifications, and certificates.
10. To track certificate validity and re-verification due dates.
11. To provide automated alerts and reminders for certificate expiry.
12. To provide role-specific dashboards for monitoring verification status and pendency.
13. To provide search, filtering, document upload, export, and printing facilities.
14. To implement role-based authentication and secure access to system resources.

4. Work Plan:-

The project will be developed as a web-based application only and will be completed within a period of four weeks.

Phase I – Requirement Analysis & System Design

- Analyze SIH26036 requirements.
- Study the verification and re-verification workflow.
- Identify system stakeholders and user roles.
- Define functional and non-functional requirements.
- Design the system architecture.
- Design the database schema and entity relationships.

- Design application workflow and role-based permissions.
- Prepare UI/UX structure and wireframes.

Phase II – Core System Development

- Develop user authentication and role-based access control.
- Develop stakeholder and user management.
- Develop instrument registration and management.
- Develop verification/re-verification application module.
- Develop application tracking and status management.
- Implement document and photograph upload functionality.

Phase III – Verification, Scheduling & Certification

- Develop verification scheduling and officer allocation.
- Develop verification and inspection module.
- Implement verification checklist and observation recording.
- Implement verification result management.
- Generate digital verification certificates.
- Implement QR-enabled certificate authentication.
- Develop certificate search and retrieval functionality.

Phase IV – Dashboard, Notifications & System Integration

- Develop role-specific dashboards.
- Implement certificate validity and expiry tracking.
- Implement automated expiry reminders and notifications.
- Develop search, filtering, export, and reporting facilities.
- Integrate all developed modules.
- Perform functional, integration, security, and usability testing.
- Fix identified errors and finalize the working prototype.

5. Methodology:-

The project will follow an Agile and iterative software development methodology. The system will be developed incrementally, with individual modules being implemented and tested before complete system integration.

Development Process:

- Requirement Analysis
- System & Database Design
- UI/UX Design
- Backend Development
- Frontend Development
- Module Integration

- Testing & Debugging
- Final Working Prototype

Major System Modules:

- Authentication and User Management
- Stakeholder Management
- Instrument Management
- Verification Application Management
- Scheduling and Allocation
- Verification and Inspection
- Digital Certificate Management
- QR-based Certificate Verification
- Document Management
- Notification and Expiry Management
- Dashboard and Analytics
- Search and Reporting
- Activity/Audit Logging

6. Organisation of Work Element:-

Work Element	Major Activities
Requirement Analysis	Problem analysis, workflow identification and requirement specification
UI/UX Design	Web page design, forms, dashboards and user workflows
Database Development	Schema design, relationships, queries and data management
Backend Development	APIs, authentication, authorization, business logic and workflows
Frontend Development	User interfaces, dashboards, forms, tables and application tracking
Verification Module	Scheduling, inspection checklist, observations and verification results
Certificate Module	Digital certificate generation, QR generation and authentication
Notification Module	Expiry tracking, reminders and notification management
Testing	Functional, integration, security and usability testing
Integration	Integration of frontend, backend, database and supporting services
Documentation	Project report and supporting technical documentation

7. Time Schedule:-

The project will be completed within four weeks. Development activities will be performed in parallel by the team members to ensure completion of all major modules within the available time.

Week	Major Activities	Expected Deliverables
Week 1	Requirement analysis, Legal Metrology workflow study, system architecture, database design, UI/UX design, project setup, authentication and RBAC	System architecture, database schema, UI structure, authentication and role management
Week 2	Stakeholder management, instrument registration, instrument management, verification/re-verification application, document upload and application tracking	Functional stakeholder and instrument modules, application submission and tracking
Week 3	Verification scheduling, officer allocation, verification/inspection module, checklist, observations, verification result, digital certificate generation and QR authentication	Complete verification workflow, digital certificate and QR verification
Week 4	Dashboards, expiry tracking, notifications, search, reports, module integration, testing, debugging and finalization	Integrated and tested working web-based prototype

8. Proposed Outcome / Findings:-

At the completion of the project, a functional web-based prototype of the proposed Legal Metrology Verification System will be developed.

- Centralized digital records of weighing and measuring instruments.
- Online stakeholder and instrument registration.
- Online verification and re-verification application submission.
- Application tracking and workflow management.
- Verification scheduling and officer allocation.
- Digital recording of inspection observations and results.
- QR-enabled digital verification certificates.
- Public certificate authentication through QR codes.
- Certificate validity and expiry monitoring.
- Automated expiry reminders and notifications.
- Role-specific dashboards for different users.
- Search and retrieval of verification records.
- Digital document and photograph management.
- Report generation, export, and printing facilities.
- Secure role-based access to system functionality.

The project will demonstrate how a centralized web-based platform can streamline the verification and certification workflow and improve accessibility, transparency, and record management.

Budget Estimates (with details) (Hardware/Software):-

1. Software Development Cost / Software Testing / Hosting / DNS etc.

Item	Estimated Cost
Software Development Tools	₹0
Programming Languages & Frameworks	₹0
Database Software	₹0
UI/UX Development Tools	₹0
Testing Tools	₹0
Cloud Hosting	₹1,500
Domain/DNS	₹1,000
Cloud/File Storage	₹1,000
Subtotal	₹3,500

The project will primarily use open-source technologies and available free/student-tier development services.

2. Report Writing

Item	Estimated Cost
Project Report Printing	₹1,000
Report Binding/Documentation	₹1,000
Subtotal	₹2,000

3. Any Other Cost Included

Item	Estimated Cost
Miscellaneous / Contingency	₹1,000
Subtotal	₹1,000

Estimated Total Project Budget: ₹6,500

Existing laboratory computers, internet facilities, and available development infrastructure will be used. No dedicated hardware is required as the proposed project is a web-based software solution.

Utilization of the Outcome of Project: -

The proposed system can be utilized as a prototype/reference implementation for digitizing the verification and certification workflow of weighing and measuring instruments.

The system can potentially be used by:

- Legal Metrology Departments
- Legal Metrology Officers (LMOs)
- Government Approved Test Centres (GATCs)
- Businesses and instrument owners
- Authorized stakeholders
- Consumers/public users for certificate verification

The proposed system can be further enhanced in the future by integrating additional government services, state-specific configurations, advanced analytics, notification gateways, and other authorized systems.

Name and address of real client if not an SIH problem: -

Not Applicable – The project is based on Smart India Hackathon Problem Statement SIH26036.

Organization: Ministry of Consumer Affairs, Food & Public Distribution

Department: Department of Consumer Affairs

Name and Signature of the Guide:- Name: _____ Signature: _____	Submitted To: Name of Lab In-Charge: _____ Designation: _____
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Submitted By:

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